



Endometriosis

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In endometriosis, functioning endometrial cells are implanted in the pelvis outside the uterine cavity. Symptoms depend on location of the implants. The classic triad of symptoms is dysmenorrhea, dyspareunia, and infertility, but symptoms may also include dysuria and dyschezia. Severity of symptoms is not related to disease stage. Diagnosis is by direct visualization and sometimes biopsy, usually via laparoscopy. Imaging studies (transvaginal ultrasound, MRI) are useful in diagnosing more advanced cases (eg, endometriosis involving the ovary [endometrioma]). Treatments include nonsteroidal anti-inflammatory drugs, medications to suppress ovarian function and endometrial tissue growth, surgical ablation and excision of endometriotic implants, and, if disease is severe and no childbearing is planned, hysterectomy alone or hysterectomy with bilateral salpingo-oophorectomy.

Endometriosis is usually confined to the peritoneal or serosal surfaces of pelvic organs, commonly the ovaries, broad ligaments, posterior cul-de-sac, uterosacral ligaments, and peritoneum.

Less common sites include the fallopian tubes, rectovaginal septum, and serosal surfaces of the small and large intestines, ureters, bladder, vagina, cervix, surgical scars, and, more rarely, the lung, pleura, and pericardium.

Bleeding from peritoneal implants is thought to initiate sterile inflammation, followed by fibrin deposition, adhesion formation, and, eventually, scarring.

Prevalences of endometriosis in a meta-analysis of 17 studies were (1)

- 18% in reproductive-aged women
- 31% in infertile women
- 42% in women with chronic [pelvic pain](#)

Diagnosis of endometriosis may be delayed. The time from onset of symptoms to diagnosis varies from 4 to 11 years (2), with the average age at diagnosis being 28 years (3). Endometriosis also occurs in adolescents and, rarely, in premenarchal girls.

References

1. [Moradi Y, Shams-Beyranvand M, Khateri S, et al](#): A systematic review on the prevalence of endometriosis in women. *Indian J Med Res.* 2021;154(3):446-454. doi:10.4103/ijmr.IJMR_817_18
2. [Arruda MS, Petta CA, Abrão MS, Benetti-Pinto CL](#): Time elapsed from onset of symptoms to diagnosis of endometriosis in a cohort study of Brazilian women. *Hum Reprod.* 2003;18(4):756-759. doi:10.1093/humrep/deg136
3. [Singh S, Soliman AM, Rahal Y, et al](#): Prevalence, Symptomatic Burden, and Diagnosis of Endometriosis in Canada: Cross-Sectional Survey of 30 000 Women. *J Obstet Gynaecol Can.* 2020;42(7):829-838. doi:10.1016/j.jogc.2019.10.038

Etiology and Pathophysiology of Endometriosis

The most widely accepted hypothesis for the pathogenesis of endometriosis is that endometrial cells are transported from the uterine cavity during menstruation and subsequently become implanted at ectopic sites. Retrograde menstruation through the fallopian tubes is common and could transport endometrial cells intra-abdominally.

Other hypotheses for the pathogenesis of endometriosis include coelomic metaplasia (transformation of peritoneal mesothelium into endometrium-like glands); Mullerian rests (endometrium-like cells develop from residual embryologic Mullerian cells); transport of endometrial cells through the lymphatic or circulatory systems (1).

The increased incidence in first-degree relatives of women with endometriosis and in large twin studies (2) suggests that heredity is a factor.

Potential risk factors for endometriosis are

- Family history of first-degree relatives with endometriosis
- Delayed childbearing or nulliparity
- Early menarche
- Late [menopause](#)
- Shortened menstrual cycles (< 27 days) with menses that are heavy and prolonged (> 8 days)
- Müllerian duct defects (eg, noncommunicating uterine horn remnant, cervical hypoplasia with obstruction of the uterine outflow tract)
- Exposure to diethylstilbestrol in utero

Potential protective factors seem to be

- Multiple births
- Prolonged lactation
- Late menarche

- Long-term use of low-dose oral contraceptives (continuous or cyclic)
- Regular exercise (especially if begun before age 15, if done for > 4 hours/week, or both)

Microscopically, endometriotic implants consist of glands and stroma histologically identical to intrauterine endometrium. These tissues contain estrogen and progesterone receptors and thus usually grow, differentiate, and bleed in response to changes in hormone levels during the menstrual cycle; also, some endometriotic implants produce estrogen and prostaglandins. Implants may become self-sustaining or regress, as may occur during pregnancy (probably because progesterone levels are high). Ultimately, the implants cause inflammation and increase the number of activated macrophages and the production of proinflammatory cytokines.

In patients with severe endometriosis and distorted pelvic anatomy, the [infertility](#) rate is high, possibly because the distorted anatomy and inflammation interfere with mechanisms of ovum pickup, oocyte fertilization, and tubal transport. Reasons for impaired fertility are unclear but may include:

- Increased incidence of luteinized unruptured ovarian follicle syndrome (trapped oocyte)
- Increased peritoneal prostaglandin production or peritoneal macrophage activity that may affect fertilization, sperm, and oocyte function
- Nonreceptive endometrium (because of luteal phase dysfunction or other abnormalities)

Etiology and pathophysiology references

1. [Burney RO, Giudice LC](#): Pathogenesis and pathophysiology of endometriosis. *Fertil Steril*. 2012;98(3):511-519. doi:10.1016/j.fertnstert.2012.06.029
2. [Saha R, Pettersson HJ, Svedberg P, et al](#): Heritability of endometriosis. *Fertil Steril* 104 (4):947–952, 2015. doi: 10.1016/j.fertnstert.2015.06.035

Symptoms and Signs of Endometriosis

Some women with extensive endometriosis are asymptomatic; some with minimal disease have incapacitating pain.

The classic triad of symptoms of endometriosis is dysmenorrhea, dyspareunia, and infertility. Cyclic midline pelvic pain preceding or during menses ([dysmenorrhea](#)) and pain in the abdomen during sexual intercourse ([deep dyspareunia](#)) are typical and can be progressive and chronic (lasting > 6 months). Dysmenorrhea that begins after several years of relatively pain-free menses is an important diagnostic clue.

Endometriosis is suspected in women with [infertility](#), particularly if accompanied by pelvic pain.

Adnexal masses (endometriomas) are also typical. Intermenstrual bleeding is possible; ovarian dysfunction can cause premature loss of estrogen and [progesterone](#) production from the ovarian follicle, leading to premature breakdown of the endometrial support.

[Interstitial cystitis](#) with suprapubic pain, dysuria, urinary frequency, and urge incontinence is common in women with endometriosis ([1](#)).

Symptoms often lessen or resolve during pregnancy. Endometriosis tends to become inactive after menopause because estrogen and progesterone levels decrease.

Symptoms and signs can vary depending on location of implants.

- Ovaries: Formation of an endometrioma (a cystic mass localized to an ovary), which occasionally ruptures or leaks, causing acute abdominal pain and peritoneal signs
- Adnexal structures: Formation of adnexal adhesions, resulting in a pelvic mass or pain
- Bladder: Dysuria, hematuria, suprapubic or pelvic pain (particularly during urination), urinary frequency, urge incontinence, or a combination
- Large intestine: Dyschezia, abdominal bloating, diarrhea or constipation, or rectal bleeding during menses
- Extrapelvic structures: Vague abdominal pain (sometimes)

Pelvic examination may be normal, or findings may include cervix deviated from midline, retroverted and fixed uterus, fixed ovarian mass, ovarian tenderness, thickened or nodular rectovaginal septum, or nodules on the uterosacral ligament. Rarely, lesions can be seen on the vulva or cervix or in the vagina, umbilicus, or surgical scars.

Symptoms and signs reference

1. [Wu CC, Chung SD, Lin HC](#): Endometriosis increased the risk of bladder pain syndrome/interstitial cystitis: A population-based study. *Neurourol Urodyn*. 2018;37(4):1413-1418. doi:10.1002/nau.23462

Diagnosis of Endometriosis

- Direct visualization, usually during pelvic laparoscopy
- Biopsy
- Sometimes pelvic ultrasound or MRI

Diagnosis of endometriosis is suspected based on typical symptoms. Misdiagnosis as [pelvic inflammatory disease](#), [urinary tract infection](#), or [irritable bowel syndrome](#) is common. Negative cervical and/or urine cultures suggest the possibility of endometriosis.

The diagnosis of endometriosis must be confirmed by direct visualization, usually via pelvic laparoscopy but sometimes via laparotomy, pelvic examination, sigmoidoscopy, or cystoscopy. Biopsy is not required, but results confirm the diagnosis.

Macroscopic appearance (eg, clear, red, blue, brown, black) and size of implants vary during the menstrual cycle. However, typically, early lesions are clear or red (hemorrhagic). As the blood in the lesions oxidizes, they turn purple, then brown; they then turn to bluish or purplish brown spots that are > 5 mm and resemble powder burns.

Microscopically, endometrial glands and stroma are usually present. Stromal elements in the absence of glandular elements indicate a rare variant of endometriosis called stromal endometriosis.

Imaging tests may not reliably detect endometriosis. However, a pelvic ultrasound or MRI showing an ovarian cyst consistent with an endometrioma is highly suggestive of the diagnosis. The presence and size of ovarian endometriomas are part of the staging system for endometriosis (stage III: small endometriomas; stage IV: large endometriomas), and a decrease in endometrioma size can show response to treatment.

Because endometrial tissue has a unique MR signal, MRI is becoming increasingly useful for evaluating patients who may have endometriosis (1). T1- and T2-weighted MRI can detect some endometriotic lesions in the pelvis, particularly larger lesions. Hemorrhage in the fallopian tubes or in an ovarian cyst without an increase in blood flow suggests endometriosis. Multiple large areas of endometriosis located in the cul de sac indicate severe (stage IV) endometriosis.

No laboratory tests contribute to the diagnosis of endometriosis, although biomarkers such as plasma microRNA are being studied in clinical trials (2).

Pearls & Pitfalls

- Consider endometriosis if patients have persistent cyclic pelvic pain, particularly if they also have dyspareunia or infertility.

Staging endometriosis helps physicians formulate a treatment plan and evaluate response to therapy. According to the American Society for Reproductive Medicine, endometriosis may be classified as stage I (minimal), II (mild), III (moderate), or IV (severe), based on

- Number, location, and depth of implants
- Presence of endometriomas and filmy or dense adhesions (see table [Stages of Endometriosis](#))

TABLE		
Stages of Endometriosis		
Stage	Classification	Description
I	Minimal	A few superficial implants
II	Mild	More and slightly deeper implants
III	Moderate	Many deep implants, small endometriomas on one or both ovaries, and some filmy adhesions

Stage	Classification	Description
IV	Severe	Many deep implants, large endometriomas on one or both ovaries, and many dense adhesions, sometimes with the rectum adhering to the back of the uterus

Other classification or clinical outcome prediction systems have been developed for endometriosis, but few have been found to be clinically useful. The endometriosis fertility index (EFI) has been found to correlate with spontaneous pregnancy rates (without use of assisted reproductive technologies) after surgery for endometriosis, however results varied across studies (3).

Diagnosis references

1. [Guerriero S, Saba L, Pascual MA, et al](#): Transvaginal ultrasound vs magnetic resonance imaging for diagnosing deep infiltrating endometriosis: systematic review and meta - analysis. *Ultrasound Obstet Gynecol* 51 (5):586–595, 2018. doi: 10.1002/uog.18961
2. [Nisenblat V, Bossuyt PM, Shaikh R, et al](#): Blood biomarkers for the non-invasive diagnosis of endometriosis. *Cochrane Database Syst Rev* 2016;2016(5):CD012179. Published 2016 May 1. doi:10.1002/14651858.CD012179
3. [Vesali S, Razavi M, Rezaeinejad M, et al](#): Endometriosis fertility index for predicting non-assisted reproductive technology pregnancy after endometriosis surgery: a systematic review and meta-analysis. *BJOG*. 2020;127(7):800-809. doi:10.1111/1471-0528.16107

Treatment of Endometriosis

- Nonsteroidal anti-inflammatory drugs (NSAIDs)
- Estrogen-progestin contraceptives
- Medications to suppress ovarian function
- Surgical resection or ablation of endometriotic tissue
- Total abdominal hysterectomy with or without bilateral salpingo-oophorectomy, if disease is severe and the patient has completed childbearing

Patients with suspected endometriosis are often treated empirically first. If symptoms are controlled with noninvasive measures, surgery can be avoided.

Symptomatic medical treatment begins with analgesics (usually NSAIDs) and hormonal contraceptives. If these are ineffective, patients are counseled about proceeding with more intensive pharmacologic treatment (GnRH analogs, aromatase inhibitors, or [danazol](#)) or undergoing laparoscopy for confirmation of the diagnosis and concomitant treatment.

Diagnostic laparoscopy is done to detect endometriosis or other etiologies of symptoms. If endometriosis is present, lesions are treated during the same procedure. Conservative surgical treatment is excision or ablation of endometriotic implants and removal of pelvic adhesions. Ovarian cystectomy can be performed if an ovarian endometrioma is present. More definitive treatment must be individualized based on the patient's age, symptoms, and desire to preserve fertility and on the extent of the disorder.

Following surgical treatment of endometriosis, hormonal contraceptives or other medications are typically given. In most patients, endometriosis recurs within 6 months to 1 year after surgery alone or when medications are discontinued, unless ovarian function is permanently and completely ablated.

Total abdominal hysterectomy with or without bilateral salpingo-oophorectomy is considered definitive treatment of endometriosis. However, endometriosis can recur even after hysterectomy in premenopausal women or those on estrogen therapy.

Pharmacologic therapy

Usually, NSAIDs are used to relieve pain. They may be all that is needed if symptoms are mild and the patient has completed childbearing.

Medications that suppress ovarian function inhibit the growth and activity of endometriotic implants. Pharmacologic treatment is effective for controlling pain but does not change fertility rates in women with minimal or mild endometriosis.

The following hormonal contraceptives are commonly used as initial therapy:

- Continuous combination (estrogen-progestin) [contraceptives](#)
- Progestins, for patients who have contraindications to estrogen therapy

The following medications are usually used only if symptoms are not well controlled with NSAIDs or hormonal contraceptives:

- Gonadotropin-releasing hormone (GnRH) agonists (eg, [leuprolide](#)) and antagonists (eg, [elagolix](#))
- Aromatase inhibitors (eg, [anastrozole](#), [letrozole](#))
- [Danazol](#)

GnRH agonists initially increase hypothalamic GnRH secretion, but continued use then temporarily decreases pituitary release of follicle-stimulating hormone (FSH), resulting in a decrease in estrogen production by the ovaries; however, treatment is limited to ≤ 6 months because long-term use may result in bone loss. If treatment lasts > 4 to 6 months, a progestin or a bisphosphonate may be used concurrently to minimize bone loss. If endometriosis recurs, women may need to be treated again.

The **GnRH antagonist** [elagolix](#) directly decreases GnRH secretion and thus suppresses pituitary release of FSH and estrogen production by the ovaries. It is available in 2 different doses; the higher dose is available to treat [dyspareunia](#) as well as other symptoms of endometriosis. Long-term use may result in

bone loss. If treatment lasts > 6 months, a progestin may be used concurrently (as add-back therapy) to minimize bone loss.

The GnRH antagonist [relugolix](#) combined with [estradiol](#) 1 mg and [norethindrone](#) 0.5 mg is undergoing clinical trials for use as primary treatment for endometriosis; this combination minimizes hot flashes and bone loss. Use is limited to 24 months because possible continued bone loss may be irreversible.

Aromatase inhibitors may be considered when GnRH analogs are ineffective in suppressing endometriosis, because some endometriosis implants have intrinsic aromatase activity and can produce estrogen from conversion of circulating androgen precursors (1).

[Danazol](#), a synthetic androgen and an antigonadotropin, inhibits ovulation. However, its androgenic adverse effects limit its use.

Cyclic or continuous **combination oral contraceptives** given after GnRH analogs or aromatase inhibitors may slow disease progression and are warranted for women who are not planning to conceive a pregnancy immediately after discontinuing those more intensive medications.

Surgery

Most women with moderate to severe endometriosis are treated most effectively by ablating or excising as many implants as possible while restoring pelvic anatomy and preserving fertility as much as possible. Superficial endometriotic implants can be ablated. Deep, extensive implants should be excised.

Specific indications for laparoscopic surgery include

- Moderate to severe pelvic pain that does not respond to medications
- Endometrioma
- Significant pelvic adhesions
- Fallopian tube obstruction
- A desire to become pregnant within a few months after surgery
- Dyspareunia (surgical treatment is second-line treatment unless performed during diagnostic laparoscopy)

Laparoscopic ablation or resection is the most common surgical procedure for an endometriotic implant; peritoneal or ovarian lesions can sometimes be electrocauterized, excised, or, uncommonly, vaporized with a laser. Endometriotic implants usually recur within 1 to 2 years without medications. Hormonal treatment of endometriosis is contraindicated during pregnancy, so patients with infertility are usually counseled to try to conceive soon after surgery.

Ovarian cystectomy is indicated if an ovarian endometrioma is present.

After laparoscopy or cystectomy, fertility rates are inversely proportional to the severity of endometriosis. If resection is incomplete, GnRH agonists are sometimes given during the perioperative

period, but whether this tactic increases fertility rates is unclear. Laparoscopic resection of the uterosacral ligaments with electrocautery or a laser may reduce midline pelvic pain.

Rectovaginal endometriosis, the most severe form of the disease, can be treated with the usual treatments for endometriosis; however, colon resection or surgery may be required to prevent obstruction of the colon.

Hysterectomy with or without bilateral salpingo-oophorectomy should usually be reserved for patients who have moderate to severe pelvic pain, have completed childbearing, and prefer a definitive procedure. Hysterectomy removes adhesions or implants that adhere to the uterus or cul-de-sac.

If women < 50 years old undergo hysterectomy with bilateral salpingo-oophorectomy, estrogen therapy should be considered (eg, to treat menopausal symptoms). Also, concomitant continuous progestin therapy (eg, [medroxyprogesterone](#) acetate 2.5 mg orally once a day) is often recommended because if estrogen is given alone, residual tissue may grow, resulting in recurrence. If symptoms persist after salpingo-oophorectomy in women > 50 years old, continuous progestin therapy alone ([norethindrone](#) acetate 2.5 to 5 mg, [medroxyprogesterone](#) acetate 5 mg orally once a day, micronized [progesterone](#) 100 to 200 mg orally at bedtime) can be tried.

Treatment reference

1. [Ferrero S, Gillott DJ, Venturini PL, Remorgida V](#): Use of aromatase inhibitors to treat endometriosis-related pain symptoms: a systematic review. *Reprod Biol Endocrinol*. 2011;9:89. Published 2011 Jun 21. doi:10.1186/1477-7827-9-89

Key Points

- Endometriosis is the presence of endometrial tissue implanted in the pelvis outside the uterine cavity, most commonly the ovaries, broad ligaments, posterior cul-de-sac, uterosacral ligaments, and peritoneum.
- The classic triad of symptoms is dysmenorrhea, dyspareunia, and infertility, but symptoms may also include dysuria and dyschezia.
- For suspected endometriosis, treat pain with analgesics (eg, NSAIDs) and hormonal contraceptives.
- If initial medications are ineffective, proceed with more intensive pharmacologic treatment (GnRH analogs, aromatase inhibitors, or [danazol](#)) or laparoscopy.
- Perform diagnostic laparoscopy to confirm the diagnosis with visual inspection; a biopsy is not mandatory but may aid in the diagnosis.
- During laparoscopy, ablate or excise as many implants as possible, lyse adhesions to restore normal pelvic anatomy, and remove endometriomas; depending on patient fertility goals, usually use medications that suppress ovarian function to inhibit the growth and activity of endometriotic implants.

- Endometriosis has a staging system based on severity; the stage does not correlate with severity of symptoms.
- Reserve hysterectomy with or without bilateral salpingo-oophorectomy for women who have completed childbearing or who prefer a definitive procedure.

Drug Information for the Topic



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